

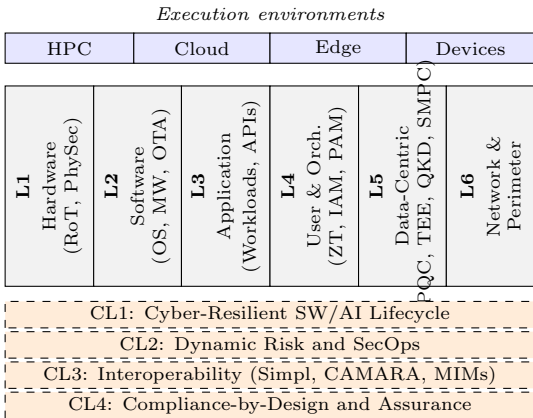


Fig. 1: Layered security model for the Cloud–Edge–IoT continuum.

	Infrastructure	Operation Domains		
HPC	Layer 1: Hardware Centric Security (Device/Edge/ Cloud/HPC)	Layer 6: Network infrastructure and perimeter security		
		Layer 5: Data Centric Security		
Cloud		Layer 2: Software Centric Security (Middleware/OS)	Layer 3: Application Centric Security (Apps/Services)	Layer 4: User Centric Security (Authentication Access/Identity / Orchestration)
Edge				
Devices				
Cross Layer 1: Automated security processes and Cyber-Resilient AI Lifecycle				
Cross Layer 2: Dynamic Risk Management & Operational Cyber Resilience				
Cross Layer 3: Interoperable Security Mechanisms				
Cross Layer 4: Agile Governance, Compliance-by-Design & Automated Assurance				

Fig. 1: Security Framework with separated security verticals in dedicated boxes.

Security Framework with Separated Security Verticals

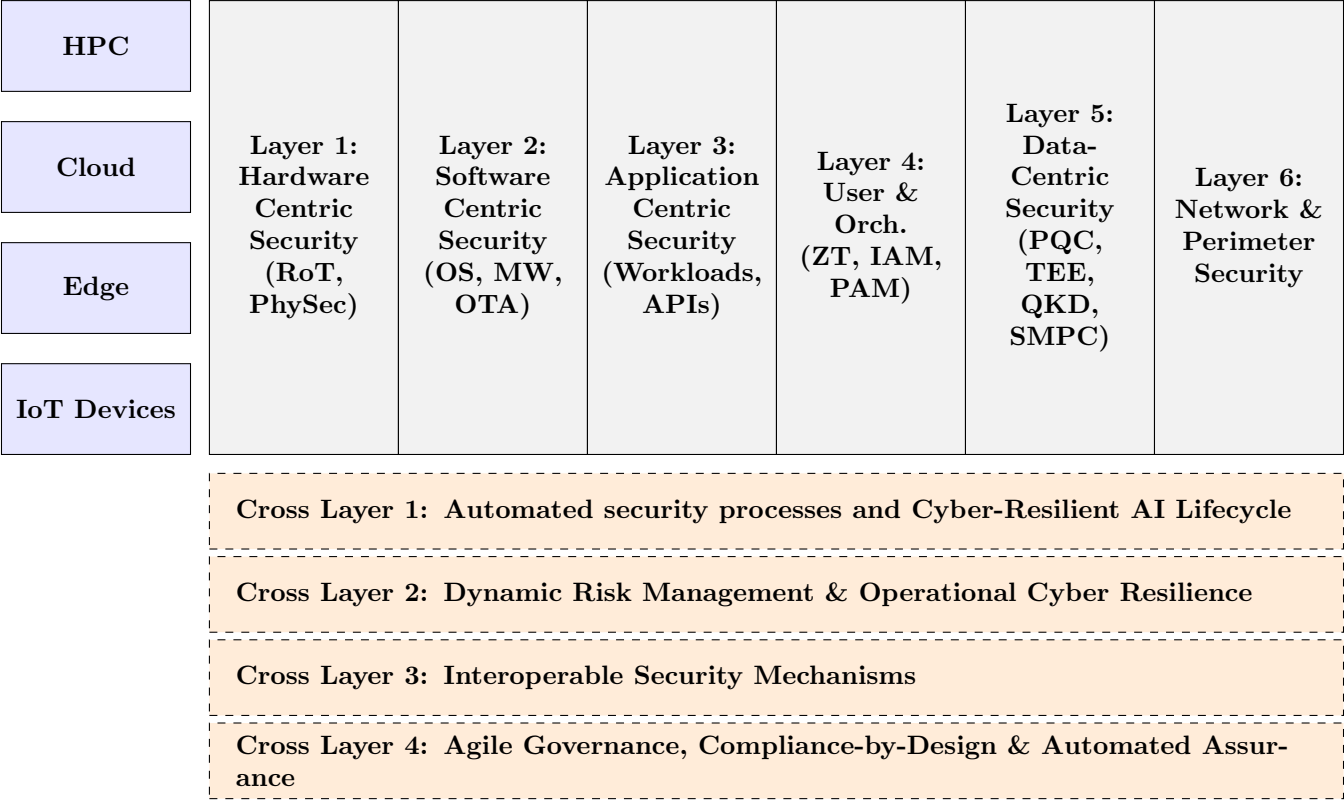
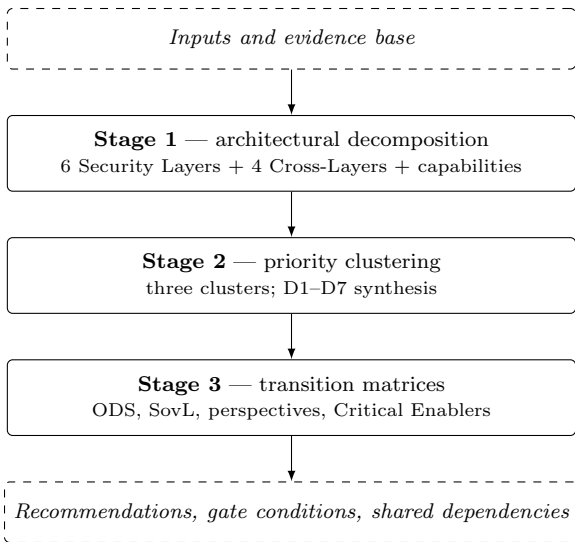


Fig. 2: Security framework with L1–L6 security layers and four cross-cutting capability domains.



Capability	C1	C2	C3	Role
L1.1 Hardware RoT	●		○	C1 primary; attestation anchor for C3
L1.2 PHY threat detection	●			C1 only
L2.1 OS/middleware	●			C1 only
L2.2 Secure OTA	●			C1 only
L4.1 Zero Trust orch.		●	●	C2/C3 shared enforcement
L4.2 Federated IAM/SSI			●	C3 only (identity primitive)
L5.1–L5.6 Data-centric	●			C1 only
L6.1 Telco-cloud/O-RAN	●	○	○	C1 primary; telemetry for C2; NaaS for C3
L6.2–L6.4 Network resil.	●			C1 only
CL1.2–CL1.4 AI lifecycle		●		C2 only
CL1.3 Runtime SBOMs	○		●	C1/C3 supply-chain bridge
CL2.1–CL2.6 SecOps/CTI		●		C2 only
CL3.1–CL3.5 Interop/fed.			●	C3 only
CL4.2 Continuous cert.			●	C3 only
CL4.4 Federated AI gov.			●	C3 only
CL4.7 AI accountability		●		C2 only

● primary cluster; ○ shared/bridge capability.

Phase 1 (2026–2030)**Phase 2 (2030–2035)****Phase 3 (2035–2040)****C1 ODS**

Applied R&D → Frag. Pilot

Standardised Integration

Ubiquitous Operation

C1 SovL

High Foreign Dep.

Emerging EU Ecosystem

Full Sovereign Autonomy

①

②

Critical Enablers: 1 = EUCC “High” for EU-origin RISC-V SE; 2 = production EU-manufactured EUCC “High” SE; ...

Figure 1: Cluster trajectories across the 2026–2040 roadmap horizon. Each cluster occupies a two-row lane: ODS (deployment maturity) above, SovL (sovereignty position) below. Numbered markers at phase boundaries identify the Critical Enabler that gates progression.